

Cohort Retention Basics

Deliverables: cohort table, heatmap, insights | Dataset: Online Retail II (UCI), 1 Dec 2009 - 9 Dec 2011

Objective

Build a cohort retention table by signup month and use it to understand how customer retention evolves over time.

Method

- **Cohort definition.** A customer's cohort is the calendar month of their first valid purchase. The dataset has no signup date, so first purchase is the standard proxy for signup.
- **Retention window.** Calendar months since the cohort month (M0 = cohort month, M1 = next month, and so on). A customer counts as retained in Mk if they placed at least one valid order in that month. Every cell uses the same definition.
- **Retention %** = customers active in Mk / cohort size (M0) x 100.
- **Partial month removed.** The data ends on 9 Dec 2011, so Dec 2011 is excluded. This keeps all windows as full calendar months and gives 24 cohorts (Dec 2009 to Nov 2011).

Data cleaning

Step	Rows remaining
Raw rows	1,067,371
After dropping missing CustomerID	824,364
After dropping cancellations (C-invoices)	805,620
After dropping Quantity<=0 / Price<=0	805,549
After excluding partial Dec-2011	788,244

Result: 5,850 unique customers across 24 cohorts. Guest checkouts (no Customer ID) are dropped because they cannot be tracked over time; cancellations (invoices starting with C) and non-positive quantity/price rows are dropped because they are not purchases.

Caveat on the first cohort

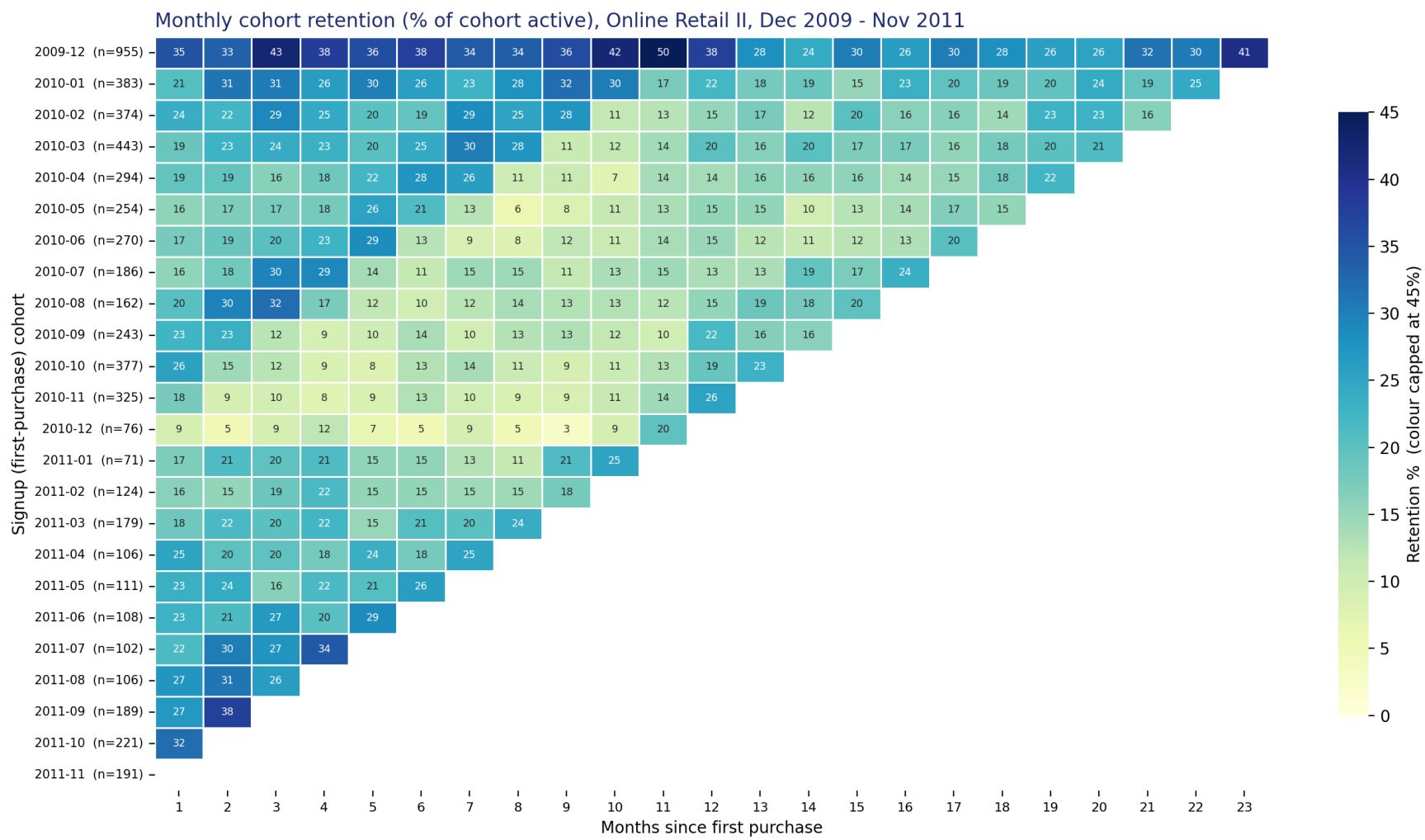
The Dec 2009 cohort (955 customers) is **left-censored**: the data starts that month, so it contains many long-standing customers, not just new ones. It retains far better than any other cohort and should not be compared to the rest. The large Jan-Mar 2010 cohorts are probably inflated the same way, since customers who bought before Dec 2009 but not in that month appear as "new".

Deliverable 1: Cohort retention table (% of cohort active)

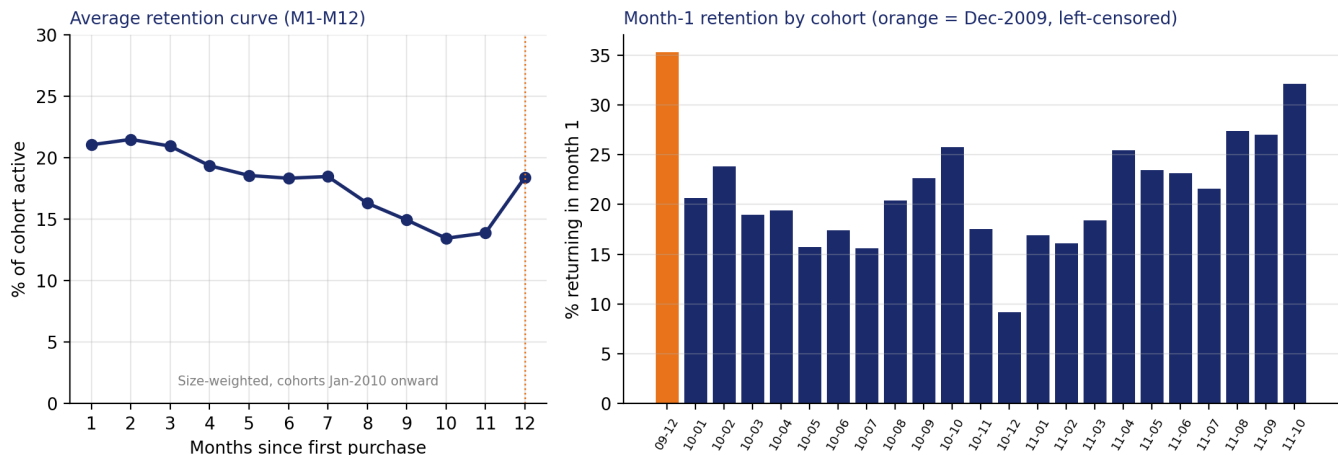
Cohort	Size	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18	M19	M20	M21	M22	M23
2009-12	955	35.3	33.4	42.5	38.0	35.9	37.7	34.2	33.6	36.2	42.2	49.5	37.6	28.3	24.4	30.3	26.3	30.3	28.3	26.0	25.5	31.5	30.5	40.7
2010-01	383	20.6	31.1	30.5	26.4	30.0	25.8	23.0	27.9	31.9	30.3	17.2	22.2	17.8	18.8	15.1	23.5	19.8	18.5	19.6	24.3	19.3	24.5	
2010-02	374	23.8	22.5	29.1	24.6	20.1	19.3	28.6	25.4	27.5	11.5	12.6	15.2	17.4	12.3	20.1	16.0	16.3	14.4	23.0	23.0	16.3		
2010-03	443	19.0	23.0	24.2	23.3	20.3	24.6	30.2	27.5	10.8	11.5	14.2	20.1	16.3	20.1	16.9	17.4	15.6	17.6	20.1	21.2			
2010-04	294	19.4	19.4	16.3	18.4	22.4	27.6	26.2	10.5	10.9	7.5	13.9	13.9	15.6	15.6	15.6	13.9	15.0	18.0	22.4				
2010-05	254	15.7	16.9	17.3	17.7	25.6	21.3	12.6	5.9	8.3	11.4	13.4	15.4	15.4	9.8	12.6	13.8	16.5	15.4					
2010-06	270	17.4	18.9	20.4	23.0	28.5	12.6	8.9	8.1	11.9	10.7	13.7	14.8	12.2	11.1	12.2	13.3	20.4						
2010-07	186	15.6	18.3	29.6	29.0	14.0	11.3	14.5	14.5	11.3	13.4	14.5	13.4	13.4	19.4	17.2	23.7							
2010-08	162	20.4	29.6	32.1	17.3	11.7	9.9	12.3	13.6	13.0	13.0	12.3	15.4	18.5	17.9	19.8								
2010-09	243	22.6	23.5	12.3	9.1	10.3	13.6	9.9	12.8	13.2	11.5	9.9	21.8	16.5	16.0									
2010-10	377	25.7	14.6	12.5	8.8	8.2	13.0	13.8	10.6	9.3	10.6	13.0	19.1	23.1										
2010-11	325	17.5	9.2	9.5	7.7	8.6	12.9	9.8	8.6	8.9	11.1	14.5	25.5											
2010-12	76	9.2	5.3	9.2	11.8	6.6	5.3	9.2	5.3	2.6	9.2	19.7												
2011-01	71	16.9	21.1	19.7	21.1	15.5	15.5	12.7	11.3	21.1	25.4													
2011-02	124	16.1	15.3	18.5	21.8	15.3	15.3	14.5	14.5	17.7														
2011-03	179	18.4	21.8	20.1	22.3	15.1	20.7	20.1	24.0															
2011-04	106	25.5	19.8	19.8	17.9	23.6	17.9	25.5																
2011-05	111	23.4	24.3	16.2	21.6	20.7	26.1																	
2011-06	108	23.1	21.3	26.9	20.4	28.7																		
2011-07	102	21.6	30.4	27.5	34.3																			
2011-08	106	27.4	31.1	26.4																				
2011-09	189	27.0	37.6																					
2011-10	221	32.1																						
2011-11	191																							
Avg*		21.0	21.5	20.9	19.3	18.6	18.3	18.5	16.3	14.9	13.4	13.9	18.4	16.9	15.8	16.2	17.4	17.2	16.9	21.2	22.8	17.8	24.5	-

Size = customers in the cohort (M0 = 100%). Mk = share of the cohort with at least one order k months later; blank = not yet observable. *Avg = size-weighted average over cohorts Jan-2010 onward (Dec-2009 excluded as left-censored); at high k it rests on only a few cohorts (M22 is a single cohort).

Deliverable 2: Retention heatmap



Deliverable 3: Insights



- **Most of the loss happens in the first month.** Only about 21% of a typical new cohort buys again in M1, so roughly four in five customers do not return the following month. After that the curve is fairly flat rather than steadily decaying: 21% at M3 and about 18% at M6.
- **Retention drifts down through year one, then rebounds around M12.** The average falls to about 13-14% at M10-M11, then rises to about 18% at M12. Each cohort's M12 lands in the same calendar months a year later, so this looks like annual repeat buying (gift-ware is seasonal) rather than a change in loyalty. Averages beyond M12 rest on few cohorts and should be read cautiously.
- **Calendar seasonality is visible as diagonal bands.** In Nov 2011, 18 of 22 comparable cohorts had a higher active share than in Oct 2011. Q4 lifts retention for every cohort at once, so cohort curves should not be compared across different calendar windows without allowing for this.
- **Newer cohorts started with better M1 retention.** The 2010 cohorts (Jan-Dec) averaged about 19% in M1, against about 23% for the Jan-Oct 2011 cohorts. Part of this is timing: the Oct 2011 cohort (32%) has its M1 in the Nov 2011 seasonal peak, and Dec 2010 (9%, only 76 customers) has its M1 in the post-holiday January.
- **Acquisition volume is uneven.** Cohorts were largest in Jan-Mar 2010 (about 375-445 each, likely inflated by pre-existing customers) and again in Sep-Nov 2011 (about 190-220 each), with a trough from Dec 2010 to Aug 2011 (71-179). Retention is best where volume is also rising, in H2 2011.
- **Dec 2009 is not a fair benchmark.** It holds 35% M1 retention and stays roughly 33-50% through year one because it is a mix of new and existing customers. Excluding it lowers the average M1 rate from about 23% to 21%.

What to do with this

Focus retention effort on the first 30 days (for example a second-purchase incentive or reminder before day 30), since that is where the biggest drop occurs. Plan Q4 campaigns knowing that reactivation is strongest then. When judging a campaign, compare cohorts at the same calendar window as well as the same M-index. Add revenue per cohort next, since the company has many wholesale buyers and customer counts alone may understate their value.

Interview questions

What is a cohort? A group of users who share a defining event in the same time period, usually the month they signed up or first purchased. Grouping this way lets you compare like with like over time instead of blending old and new customers.

Why is cohort retention useful? It separates real changes in loyalty from changes in customer mix. A blended retention figure can look stable or improve just because many new customers arrived. Cohort tables show whether each new group retains better or worse than earlier groups, when customers drop off, and whether a product, pricing, or onboarding change actually worked.

Appendix A: SQL (SQLite dialect)

Run on a cleaned transactions(Invoice, CustomerID, InvoiceDate) table (cleaning steps as on page 1). Output was cross-checked against the pandas result and matches cell for cell.

```
WITH monthly AS (
    SELECT DISTINCT CustomerID,
           strftime('%Y-%m', InvoiceDate) AS ActiveMonth
    FROM transactions
),
cohorts AS (
    SELECT CustomerID, MIN(ActiveMonth) AS CohortMonth
    FROM monthly GROUP BY CustomerID
),
activity AS (
    SELECT c.CohortMonth,
           (CAST(substr(m.ActiveMonth,1,4) AS INT) - CAST(substr(c.CohortMonth,1,4) AS INT)) * 12
           + (CAST(substr(m.ActiveMonth,6,2) AS INT) - CAST(substr(c.CohortMonth,6,2) AS INT)) AS MonthIndex,
           m.CustomerID
    FROM monthly m JOIN cohorts c USING (CustomerID)
),
counts AS (
    SELECT CohortMonth, MonthIndex, COUNT(DISTINCT CustomerID) AS Customers
    FROM activity GROUP BY CohortMonth, MonthIndex
)
SELECT a.CohortMonth, a.MonthIndex, a.Customers,
       ROUND(100.0 * a.Customers / b.Customers, 1) AS RetentionPct
FROM counts a
JOIN counts b ON b.CohortMonth = a.CohortMonth AND b.MonthIndex = 0
ORDER BY a.CohortMonth, a.MonthIndex;
```

Appendix B: Python (pandas) core

```
df["Month"] = df.InvoiceDate.dt.to_period("M")
act = df[["CustomerID", "Month"]].drop_duplicates() # one row per customer-month
act["Cohort"] = act.groupby("CustomerID").Month.transform("min") # first purchase month
act["Idx"] = (act.Month.dt.year - act.Cohort.dt.year)*12 + (act.Month.dt.month - act.Cohort.dt.month)
counts = act.groupby(["Cohort", "Idx"]).CustomerID.nunique().unstack()
retention = counts.div(counts[0], axis=0) * 100
sns.heatmap(retention.iloc[:,1:], annot=True, fmt=".0f", cmap="YlGnBu", vmin=0, vmax=45)
```